

## SUMMARY

---

Ph.D. candidate (Computer Science) specializing in cybersecurity research, focusing on protocol security, fuzzing, and vulnerability detection. Industry experience at Apple as an Intern and Agoda (A Booking Holdings company) as a Senior Software Engineer shipping production software and automation. Strong in Python/C++/Swift, debugging, distributed pipelines (Kafka/ELK/Docker), and CI/CD. Seeking Software Engineering or Research Intern roles in Systems, Backend, Security, or Infrastructure.

## EDUCATION

---

- **State University of New York at Stony Brook** New York, United States  
• *Ph.D. Candidate in Computer Science* Fall 2023 – Present
- **Coursework:** Systems Security, Network Security, Algorithms, Big Data, Data Science, Visualization

## SKILLS

---

- **Languages:** Python, C++, Swift, Java
- **Systems & DevOps:** iOS, Git, Docker, GitHub Actions, CI/CD
- **Data/Infra & Observability:** Kafka, Elasticsearch, Logstash, NoSQL, REST APIs, osquery, eBPF

## PROJECTS

---

- **Black-Box Resiliency Analyzer for 5G Core:** Built a adaptive fuzzing and resiliency testing tool for 5G Core networks that mutates NGAP messages, exercises stateful scenarios, and flags/classifies bugs based on system behavior and responses.
- **CI/CD Pipeline Security Analysis:** Built a DevSecOps monitoring pipeline for GitHub Actions by collecting host/workflow telemetry with osquery and **eBPF**, streaming events via Kafka, and indexing with Logstash/Elasticsearch to support detection, auditing, and incident triage.
- **Malicious Repository Detection (Big Data):** Implemented scalable data pipelines and detection heuristics to identify potentially malicious repositories using big-data techniques.
- **NYC Motor Vehicle Collision Analysis:** Developed a data analytics and visualization pipeline for NYC vehicle collision datasets using Python (data cleaning/feature engineering) and D3.js (interactive dashboards) to surface trends, hotspots, and risk patterns.
- **Constraint-Based Text Generation (Gatsby Without “e”):** Designed an NLP generation pipeline using LLMs to produce constrained text outputs (character-level restrictions), including data preprocessing, prompt/constraint enforcement, and automated evaluation of generated results.

## EXPERIENCE

---

- **Apple** California, USA  
• *SWE Intern* May 2025 - Aug 2025
  - **Accessibility:** Built an accessibility feature improving usability for customers with limited mobility; shipped a production-ready change (design, implementation, testing).
- **Agoda** Bangkok, Thailand  
• *Senior Software Engineer (iOS)* Jan 2022 - June 2023
  - **Shopping Cart:** Designed and delivered a shopping-cart and multi-product checkout module using modular architecture patterns; improved checkout conversion by 15%.
  - **Multiproduct Booking:** Built booking workflows and integrated with APIs and analytics, improving reliability of the purchase funnel and reducing user friction.
  - **Analytics Platform:** Migrated analytics components from C# to a native implementation, improving data correctness and processing efficiency.

## PUBLICATION

---

- **ACM WiSec 2025:** Performance and Consistency Trade-off of the eSIM M2M Remote Provisioning Protocol.  
<https://dl.acm.org/doi/abs/10.1145/3734477.3734712>